

Echo Protocol: A Prompt-Native Persistent State Architecture for Large Language Models

An In-Context State Machine Validated Through Interactive Narrative Stress-Testing

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Independent Research · Development began May 23, 2025

Abstract

Large language models are stateless by design: all context is discarded at session end. Existing solutions such as Retrieval-Augmented Generation (RAG) address knowledge retrieval but do not provide structured, persistent world state across sessions. This paper presents Echo Protocol, a prompt-only architecture that enables any LLM to maintain structured state, temporal logic, relationship tracking, and cross-session continuity — without backend infrastructure, vector databases, or fine-tuning of any kind. The system was developed by a single researcher beginning May 23, 2025, and was stress-tested using interactive narrative roleplay as a demanding validation environment requiring simultaneous temporal reasoning, NPC relationship management, and long-horizon memory. Preliminary results on a free-tier model demonstrate functional persistent memory across 107 in-world days. This work is presented as an early-stage finding requiring broader community validation. The core design insight — that LLMs function simultaneously as language models and state processors — underlies all architectural decisions.

1. Introduction

The inability of large language models to retain structured state across sessions represents one of the most significant practical limitations of current AI systems. Every conversation begins from zero. Relationships, timelines, world states, and accumulated context vanish when a session ends. This is not a failure of model capability — it is a consequence of stateless architecture by design.

Current approaches address this limitation through external infrastructure: vector databases store and retrieve past interactions, backend services manage memory pipelines, and fine-tuning encodes domain knowledge into model weights. These solutions are effective but carry significant costs in infrastructure, latency, and accessibility.

This paper presents a different approach, grounded in a reframing of what an LLM is.

1.1 The Core Design Insight

Most practitioners approach LLMs as either a human-like conversational agent or a computational system. Those who treat them as conversational agents focus on prompting and persuasion. Those

who treat them as computational systems build infrastructure around them. Echo Protocol emerges from treating an LLM as both simultaneously.

An LLM has the natural language comprehension of a human reader and the deterministic rule-following capacity of a state machine, when instructed correctly. This dual nature means that structured state — normally requiring code and databases — can be encoded directly in natural language prompts and maintained through structured output constraints. No external system is required.

This insight is not merely philosophical. It is the engineering foundation that makes a prompt-only persistent memory architecture possible.

1.2 Scope and Claims

This paper makes the following claims, with the limitations described in Section 4:

- Echo Protocol demonstrates functional persistent state in LLMs using prompt instructions alone, with no backend infrastructure.
- The architecture is model-agnostic: it functions on any LLM capable of following structured output instructions.
- Interactive narrative roleplay serves as a demanding stress-test environment, not the intended end application.
- The system's theoretical ceiling scales directly with model context window size, at zero additional infrastructure cost.
- A Dual-Agent External Memory extension is proposed as a path to removing the remaining context boundary constraint. This extension has not yet been implemented.

2. Background and Related Work

2.1 The Statefulness Problem

Decoder-only transformer models process each forward pass independently. All representations are discarded after generation. This is a fundamental architectural property, not a bug. The consequence for deployed systems is that any continuity across sessions must be engineered externally (Jeong, 2026; Packer et al., 2023).

Injecting raw conversation history into context naively leads to rapidly growing input sizes, increasing cost quadratically and causing attention degradation on middle-range content — a phenomenon documented as 'lost in the middle' (Liu et al., 2023). Research further confirms that memory system performance is determined by structural quality, not data volume (Memori, 2026).

2.2 Existing Solutions and Their Constraints

Retrieval-Augmented Generation (RAG) retrieves semantically similar documents at inference time (Lewis et al., 2020). It addresses knowledge access but not structured state: RAG operates on semantic similarity, not chronological or relational logic. It cannot calculate that a character has aged two years, or that a relationship has evolved through a specific sequence of events.

MemGPT (Packer et al., 2023) implements OS-inspired virtual context management with external memory tiers. Mem0 (Chhikara et al., 2025) provides a production memory layer using vector storage and LLM extraction pipelines. Both require dedicated infrastructure and are not accessible to users without technical resources or API budgets.

Echo Protocol differs from all existing approaches in one critical respect: it requires nothing beyond the model and a prompt file.

3. System Design: Echo Protocol

3.1 Architecture Overview

Echo Protocol encodes a complete state management system within a single structured prompt. The architecture has three functional layers:

Schema Layer. The prompt itself, defining AI behavioral rules, output format constraints, memory encoding specifications, and hallucination prevention mechanisms. It functions as a compiler specification: the model executes it, not merely reads it.

Tracker Layer. A compressed structured state block appended to every model response. It encodes world state, character data, NPC relationships with emotional valence, active timers, event flags, and context utilization metrics in a single-line format designed for efficient attention weighting.

Active Context. The current session conversation window, which the model processes alongside the Schema and the most recent Tracker state.

3.2 The Tracker Format

The Tracker uses a compressed key-value encoding with three verbosity tiers:

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[W:world][P:char/age/s:LT>ST/days][PC:status][I:items][O:goal][Rp:faction]
[T:threat][L:location][N:npc/age/role/m:mood|r:relation|G:goal|K:skills]
[X:R:rumors|T:timer/Nd|F:flags|C:countdown|M:npc:event/level]
[CT:XX%][TK:XXk][Chk:a5f3b][E:errorcode]
```

Verbosity scales automatically with context utilization: Normal below 50%, Compact between 50-80%, Ultra-Compact above 80%. This auto-compression extends effective session length without any user intervention.

3.3 Key Innovations

Dynamic Decay System. NPC relationships and world events decay over in-world time using configurable rates: standard decay, slow decay (3x), or permanent (!). This models episodic memory degradation rather than treating all past events as equally salient.

Stateful Temporal Logic. Time-skips, aging, relationship evolution, and project completion are computed directly from Tracker state — not retrieved from history. The model calculates forward from current state rather than searching backward through context.

PC Behavior Logging. Character speech pattern evolution is tracked over in-world time (Long-Term vs Short-Term), enabling authentic behavioral drift across extended sessions.

Self-Correcting Error Recovery. A typed error code system ([E:A1], [E:F2], etc.) automatically adjusts model caution level after violations, without human intervention.

Cross-Session Continuity. Resuming a session requires only pasting the last Tracker into a new conversation. The model reconstructs full world state from the Tracker alone.

Model Agnosticism. The protocol has been tested on DeepSeek, Claude, and ChatGPT. No model-specific dependencies exist.

3.4 Why This Works: The Dual-Nature Model

The architecture functions because it leverages both capacities of an LLM simultaneously. The Schema uses natural language to specify behavior — exploiting the model's language comprehension. The Tracker uses structured symbolic encoding — exploiting the model's capacity to follow deterministic output rules. Neither alone is sufficient. Together, they create a system that is both semantically aware and structurally disciplined.

This is the practical consequence of the design insight stated in Section 1.1: treating the model as human and computer simultaneously.

4. Evaluation

4.1 Why Interactive Narrative as a Stress Test

Interactive narrative roleplay was selected as the validation environment not because it is the intended application of Echo Protocol, but because it is among the most demanding environments for persistent state management.

A fully functional roleplay session requires simultaneous maintenance of: world state and geography, character relationships with emotional history, temporal consistency across days, months, or years of in-world time, project timelines and deadlines, NPC behavioral consistency with established personality, and event causality across hundreds of turns. Any memory system that passes this test passes a strict multi-dimensional evaluation that most benchmarks do not capture.

4.2 Preliminary Results

The following results were obtained from a single extended test session using DeepSeek (free tier) — a lightweight model chosen specifically to demonstrate that performance is not dependent on model scale. These results represent a conservative lower bound.

- Session duration: 107 in-world days with full narrative continuity
- Context utilization at session end: 47% (approximately 189,000 tokens)
- Active NPC tracking: 8+ characters with relationship states, mood tracking, goal persistence, and knowledge attributes
- Timer accuracy: Multiple countdown events tracked simultaneously across days
- Auto-compact: Triggered correctly at 50% and 80% thresholds
- Cross-session continuity: Verified via Tracker paste-and-resume protocol

4.3 Projected Capacity

Based on observed consumption rates of approximately 0.2-0.25% context per turn:

- Without reset: approximately 400-500 turns before reaching the 90% threshold
- With the built-in RESET CONTEXT protocol: 800+ turns with maintained continuity
- On models with 1M+ token context windows: estimated 4,000-8,000+ turns

Critically, capacity scales linearly with context window size at zero additional cost. This means every advance in model context length directly benefits Echo Protocol users without any modification to the system.

4.4 Limitations of Current Testing

The following limitations must be stated clearly. This research was conducted by a single developer with limited time. The testing represents a short-run evaluation that has not approached the system's theoretical limits. Specifically:

- Testing was conducted on a single model (DeepSeek). Cross-model systematic evaluation has not been performed.
- Only one use case domain (interactive narrative) has been formally tested. Generalization to other domains is proposed but unvalidated.
- The Dual-Agent External Memory extension (Section 5.2) has not been implemented or benchmarked.
- Long-run error accumulation behavior has not been characterized.

This work is presented as a proof-of-concept and early-stage finding. Community validation is essential and actively invited.

5. Generalization Beyond Narrative

5.1 The General Case

The Tracker architecture is domain-agnostic. Any system requiring persistent structured state, temporal logic, and cross-session continuity is a candidate application. The roleplay validation environment tested these properties under adversarial conditions. Three illustrative non-narrative domains are described below.

Industrial Automation and Control.

The Tracker's state transition model maps directly to industrial control logic. Sensor readings, actuator states, system warnings, and operational modes can be encoded as Tracker fields. The model then functions as a flexible logic controller — capable of reasoning about novel fault conditions not anticipated in hardcoded rules, unlike traditional PLC Ladder Logic. The state transition equation $S(t+1) = f(S(t), I(t))$ describes both systems identically.

Complex Game Logic and Simulation.

Resource management, faction relationships, technology progression, and time-skips in strategy or simulation games require exactly the capabilities Echo Protocol provides. The model can compute aggregate state changes across time skips directly from Tracker fields, eliminating the need to simulate tick-by-tick.

Stateful Personal Assistant.

Long-term goal tracking, project management, and relationship-aware assistance require persistent state that current conversational AI systems do not maintain. Echo Protocol provides this capability without cloud storage or external services.

5.2 Proposed Extension: Dual-Agent External Memory

The primary remaining limitation of Echo Protocol is the context window boundary. The following architecture is proposed to address this. It has not been implemented and is presented as a research direction.

A lightweight secondary model (e.g., Llama 3.2 3B) operates as a dedicated Memory Manager, receiving Tracker output from each session, writing it to local storage, and injecting relevant state into the primary model's context on demand:

Primary Model (inference) <-> Memory Manager (lightweight secondary model) | Local Storage (Tracker files)

This extension would: eliminate context window constraints entirely, remove the need for human Tracker management between sessions, enable a validation layer to prevent error accumulation, and run entirely on local hardware with open-source models at zero infrastructure cost.

6. Discussion

6.1 On Accessibility

The most significant practical consequence of a prompt-only architecture is accessibility. RAG and pipeline-based memory systems require API access, cloud infrastructure, and technical implementation capacity. Echo Protocol requires none of these. A user with access to any LLM interface — including free-tier services — can deploy the full system by pasting a single file.

This democratizes persistent AI memory in a meaningful sense: it becomes available to individuals and communities who lack engineering resources or subscription budgets.

6.2 On the Development Process

Echo Protocol was not designed from a theoretical framework. It emerged from iterative practical problem-solving: observing where AI systems failed, identifying the structural cause, and testing solutions. The system reached its current form through incremental refinement from V1 to V7 across approximately eight days.

This development path is worth noting because it illustrates that fundamental architectural insights about LLM systems can emerge from applied exploration without prior academic background in the field. The core insight — that LLMs can function as state processors when prompted correctly — was discovered through practice.

6.3 Security Considerations

Persistent memory systems introduce attack surfaces that stateless systems do not have. Memory poisoning — the deliberate injection of false state into long-term memory — represents a meaningful threat vector (MemoryGraft, 2025). OWASP's 2026 Agentic AI Top 10 identifies Memory and Context Poisoning (ASI06) as a first-class concern. Echo Protocol's Schema layer and Chk validation field provide partial structural mitigation, but formal security analysis has not been conducted.

7. Limitations and Future Work

The following represent the primary open questions requiring further investigation:

- Systematic cross-model evaluation on standardized memory benchmarks (LoCoMo, MemoryAgentBench)
- Implementation and benchmarking of the Dual-Agent External Memory extension
- Formal evaluation in non-narrative application domains
- Long-run error accumulation characterization and mitigation strategies
- Security hardening of the Tracker validation layer against poisoning attacks
- Community replication across diverse models, context sizes, and use cases

8. Conclusion

Echo Protocol demonstrates that persistent, structured memory for large language models does not require backend infrastructure, vector databases, or model fine-tuning. Through prompt engineering alone — grounded in the insight that LLMs function simultaneously as language models and state processors — a single researcher achieved cross-session state persistence, dynamic relationship tracking, temporal decay simulation, and automatic context management using only free AI tools.

The system's capacity scales directly with model context window size, meaning architectural advances in the broader field translate immediately into Echo Protocol capability gains without any system modification.

This work is presented as an early-stage finding. The results are preliminary, the testing scope is limited, and many open questions remain. Community validation, replication, and extension are not merely welcomed — they are necessary for this research to reach its potential.

Echo Protocol V7 is released as open documentation. The complete prompt file is available for immediate use, replication, and critique.

References

- Chhikara et al. (2025). Mem0: Building production-ready AI agents with scalable long-term memory. arXiv.
- Jeong, H. (2026). Trained Persistent Memory for Frozen Decoder-Only LLMs. arXiv:2603.22329.
- Lewis, P. et al. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. NeurIPS 2020.

Liu, N. et al. (2023). Lost in the Middle: How Language Models Use Long Contexts. arXiv:2307.03172.

MemMachine (2026). A Ground-Truth-Preserving Memory System for Personalized AI Agents. arXiv:2604.04853.

Memori (2026). A Persistent Memory Layer for Efficient, Context-Aware LLM Agents. arXiv:2603.19935.

MemoryGraft (2025). Persistent Compromise of LLM Agents via Poisoned Experience Retrieval. arXiv:2512.16962.

OWASP Foundation (2026). Agentic AI Security Top 10: ASI06 — Memory and Context Poisoning.

Packer, C. et al. (2023). MemGPT: Towards LLMs as Operating Systems. arXiv:2310.08560.

Saisalam, A. (2025). Echo Protocol V1-V7. Independent development. Thailand. Development began May 23, 2025.